TECH STAR SUMMIT 2025

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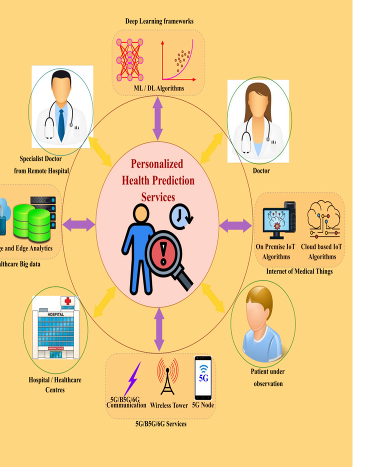
SIMATS ENGINEERING

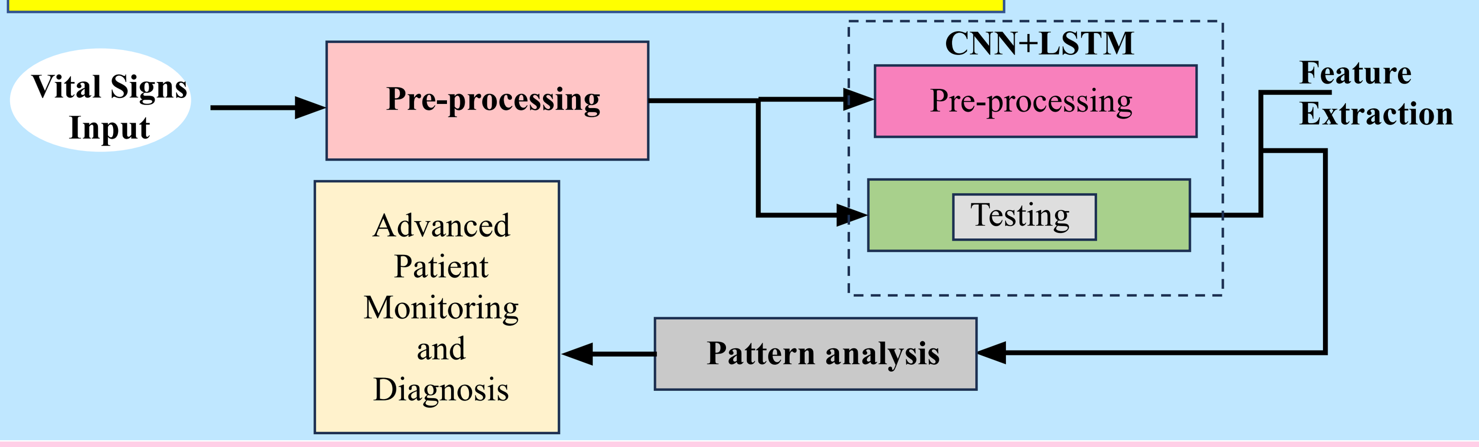
Enhancing Vital Signs Analysis and Patient Health Prediction using CNN-LSTM as an Alternative to Long Short-Term Memory(LSTM)

INTRODUCTION

- Early and accurate analysis of patient vital signs is crucial for timely intervention and effective healthcare management.
- The aim of this study is to anticipate the performance of CNN-LSTM as an alternative to LSTM models in predicting patient health status based on vital signs, to achieve improved accuracy.
- platforms like IEEE Xplore, Google Scholar, were reviewed to study patient health prediction methods using deep learning.
- CNN-LSTM demonstrates better extraction and temporal pattern recognition capabilities than standalone LSTM models.



MATERIALS AND METHODS



RESULTS

	Group	N	Mean	Std.Deviation	Std.Error Mean
Accuracy	CNN-LSTM	10	90.46	3.22372	1.01943
	LSTM	10	83.72	4.26735	1.34945

Table 1- The Mean, Standard Deviation, Standard Error Mean with an accuracy rate comparison of CNN-LSTM and LSTM.

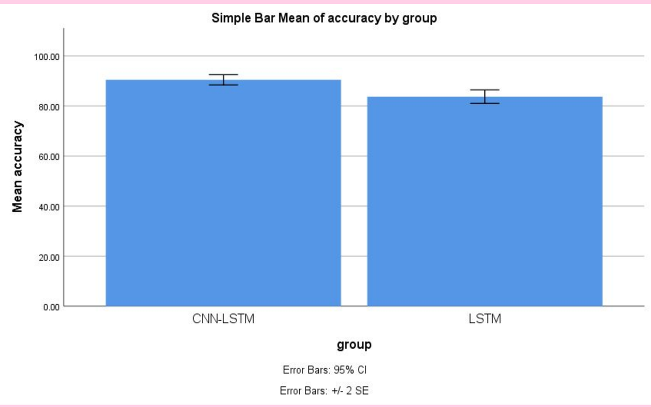


Fig.1- It depicts that the proposed algorithm gives more accuracy than compared with the LSTM.

DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.719 (independent sample T - test p>0.05) is obtained and shows that there is a statistical insignificant difference between the group 1 and group 2.
- For both CNN-LSTM and LSTM N, Mean, Std.Deviation, Std.Error Mean values are calculated.
- Overall , the accuracy of the CNN-LSTMcClassifier is 90.46% and its better than the LSTM algorithm is 83.72%.
- From the work , it is concluded that the CNN-LSTM hybrid model attains the high accuracy when comparing with LSTM model in Enhancing Vital Signs Analysis and Patient Health Prediction.

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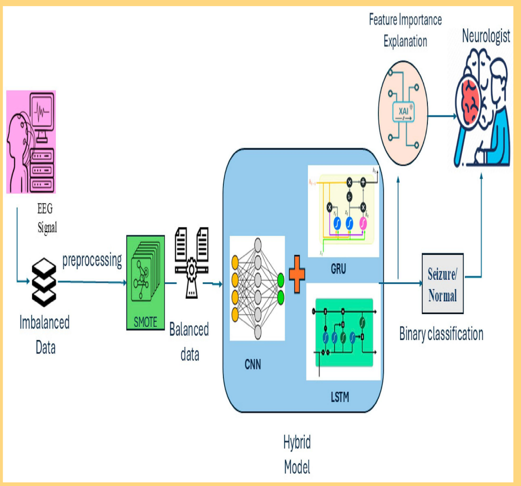
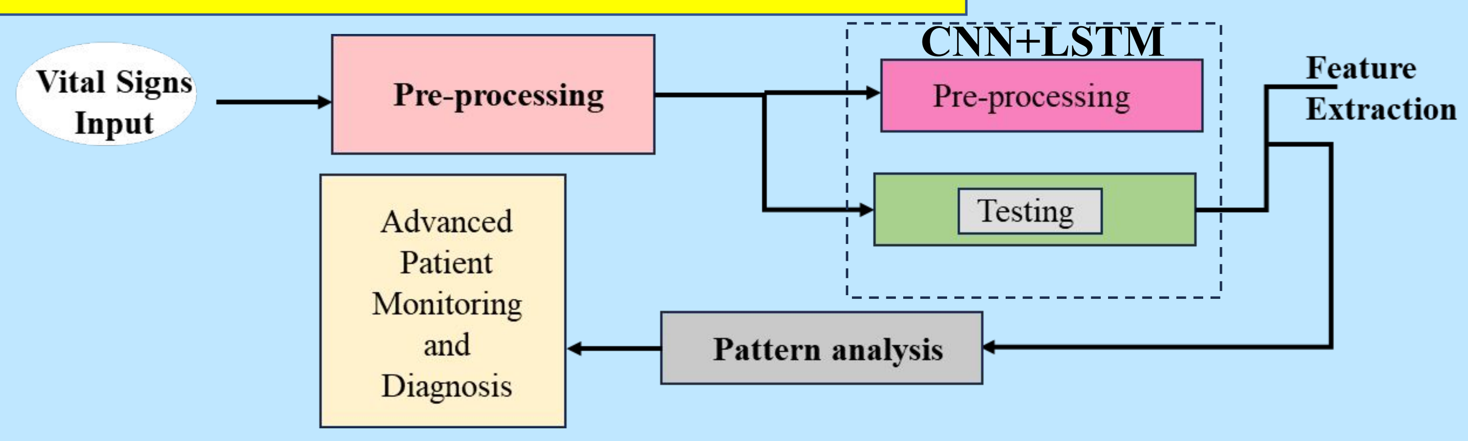
An Analysis of CNN-LSTM and GRU Models in Comparison for Improved Patient Health Monitoring and Forecasting

INTRODUCTION

- Timely monitoring of patient health is essential for effective medical care.
- This study compares CNN-LSTM and GRU models for predicting patient conditions using vital sign data.
- Research from platforms like IEEE Xplore and Google Scholar highlights deep learning's role in healthcare forecasting.
- CNN-LSTM offers strong pattern recognition, while GRU provides faster, lightweight performance.



MATERIALS AND METHODS



RESULTS

	Group	N	Mean	Std.Deviation	Std.Error Mean
Accuracy	CNN-LSTM	10	90.46	3.22372	1.01943
	GRU	10	76.77	3.06003	0.96767

Table 1- The Mean, Standard Deviation, Standard Error Mean with an accuracy rate comparison of CNN-LSTM and GRU.

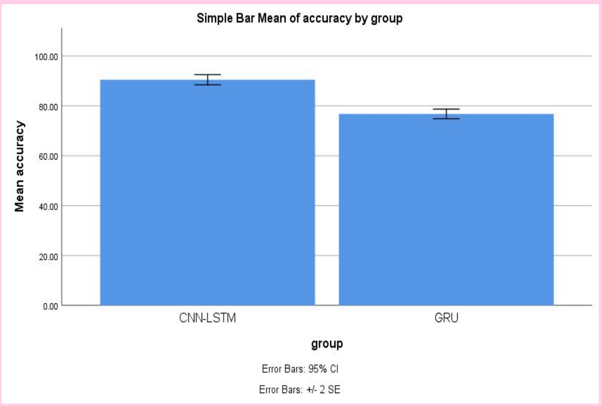


Fig.1- It depicts that the proposed algorithm gives more accuracy than compared with the GRU.

DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.785 (independent sample T - test p>0.05) is obtained and shows that there is a statistical insignificant difference between the group 1 and group 2.
- For both CNN-LSTM and GRU N, Mean, Std.Deviation, Std.Error Mean values are calculated.
- Overall , the accuracy of the CNN-LSTM model is 90.46% and its better than the GRU model is 76.77%.
- From the work , it is concluded that the CNN-LSTM hybrid algorithm attains the high accuracy when comparing with algorithm GRU in Patient Health Monitoring and Forecasting.

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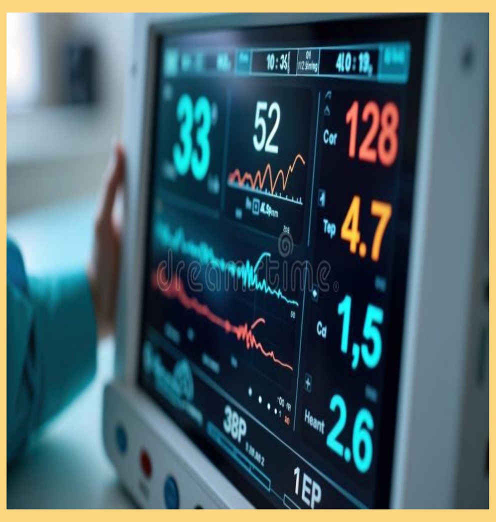
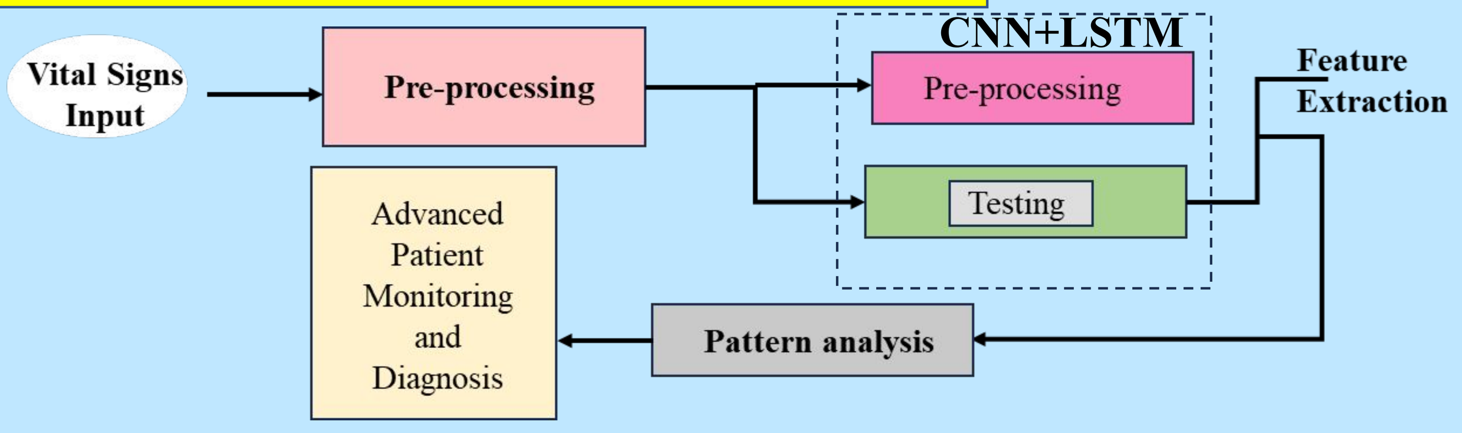
Assessing the Effectiveness of CNN-LSTM in Comparison with Transformer Models for Patient Health Forecasting

INTRODUCTION

- This study evaluates the performance of CNN-LSTM and Transformer models in analyzing vital signs for health prediction.
- CNN-LSTM is known for its ability to capture spatial-temporal features, while Transformers excel at modeling long-range dependencies.
- The comparison aims to determine the most suitable model for reliable and real-time patient health forecasting.



MATERIALS AND METHODS



RESULTS

	Group	N	Mean	Std.Deviation	Std.Error Mean
Accuracy	CNN-LSTM	10	90.46	3.22372	1.01943
	TM	10	78.93	3.04747	0.96369

Table 1- The Mean, Standard Deviation, Standard Error Mean with an accuracy rate comparison of CNN-LSTM and Transformer Models.

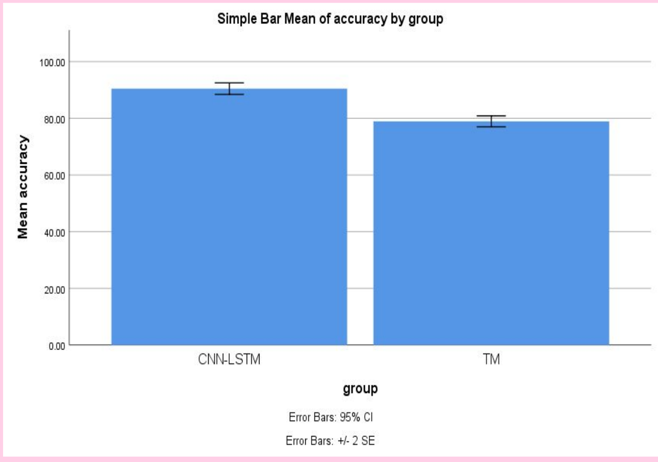


Fig.1- It depicts that the proposed algorithm gives more accuracy than compared with the TM.

DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.757 (independent sample T - test p>0.05) is obtained and shows that there is a statistical insignificant difference between the group 1 and group 2.
- For both CNN-LSTM and TM N, Mean, Std.Deviation, Std.Error Mean values are calculated.
- Overall , the accuracy of the MCNN-LSTM model is 90.46% and its better than the Transformer model is 78.93%.
- From the work , it is concluded that the CNN-LSTM model algorithm attains the high accuracy when comparing with algorithm Transformer in Patient Health Forecasting.

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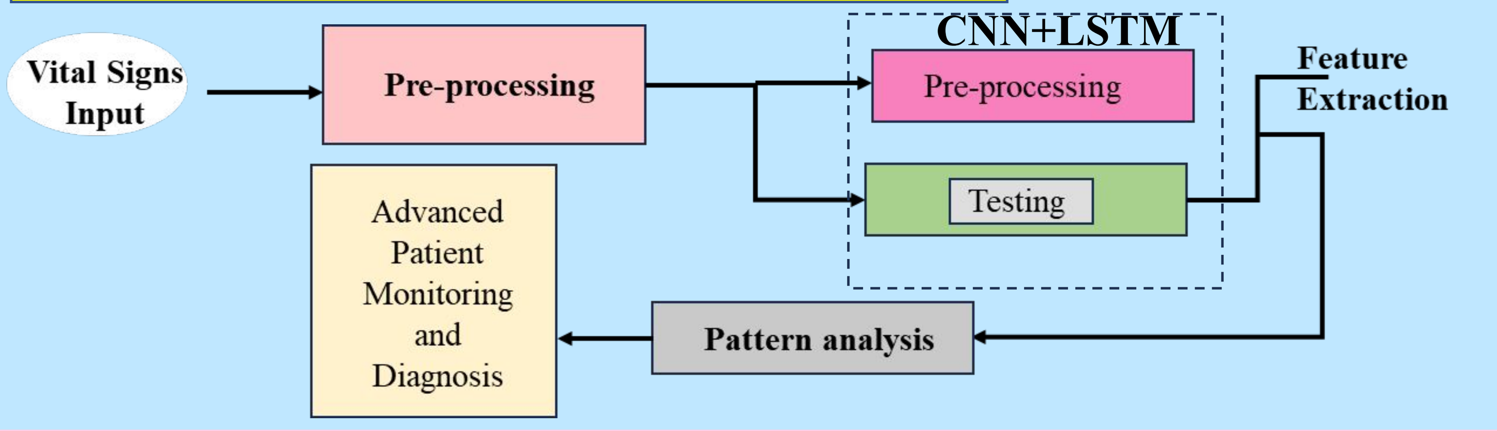
Improving Vital Signs Monitoring using CNN-LSTM and BiLSTM to Improve Patient Health Prediction

INTRODUCTION

- Effective monitoring of vital signs is essential for early detection of health risks.
- This study explores the use of CNN-LSTM and BiLSTM models to enhance the accuracy of patient health prediction.
- IEEE Xplore and Google Scholar guided the analysis of deep learning methods in healthcare.
- CNN-LSTM aids in feature extraction, while BiLSTM captures forward and backward temporal patterns.

MATERIALS AND METHODS



RESULTS

	Group	N	Mean	Std.Deviation	Std.Error Mean
Accuracy	CNN-LSTM	10	90.46	3.22372	1.01943
	BiLSTM	10	81.39	7.45879	2.35868

Table 1- The Mean, Standard Deviation, Standard Error Mean with an accuracy rate comparison of CNN-LSTM and BiLSTM .

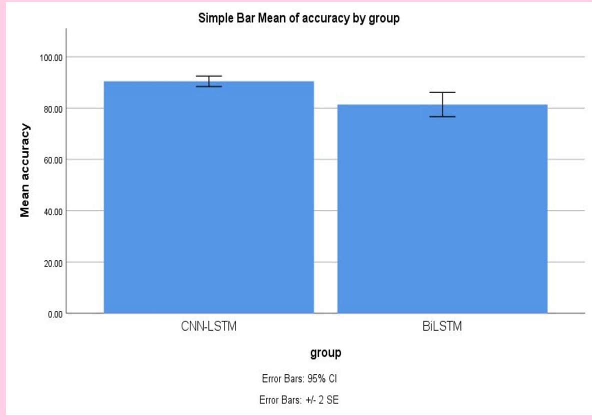


Fig.1- It depicts that the proposed algorithm gives more accuracy than compared with the BiLSTM.

DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.962 (independent sample T - test p>0.05) and there is a statistical insignificant difference between the group 1 and group 2.
- For both CNN-LSTM and BiLSTM N, Mean, Std.Deviation, Std.Error Mean values are calculated.
- Overall , the accuracy of the CNN-LSTM model is 90.46% and its better than the BiLSTM algorithm is 81.39%.
- From the work , it is concluded that the CNN-LSTM hybrid algorithm attains the high accuracy when comparing with algorithm BiLSTM to Improve Patient Vital signs monitoring.

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